Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
D		2603012

My Gain Is Not Your Loss: Dynamic Management Based on MPC and Game Theory

Abstract

The core challenge of professional sports management lies in achieving synergistic development between winning and profitability under dynamic constraints. This study takes the **New York Knicks** (from NBA) as the research object and constructs an intelligent decision-making system, aiming to transform dynamic operations into a quantifiable and adjustable scientific process through systematic modeling.

For team strength evaluation, we propose a **structured vector-based quantification method**, integrating multi-dimensional player data into a unified athletic capability vector through Entropy Weight Method and aging curves, providing a benchmark for dynamic optimization. On this basis, a **Stochastic Model Predictive Control (SMPC)** framework is established, defining the team's state across four dimensions: **competitive assets, brand value, financial reserves, and liabilities**. Through *receding horizon optimization*, the framework simultaneously optimizes cash flow, brand growth, and roster competitiveness while adhering to hard constraints such as salary caps.

Player recruitment is modeled as a high-dimensional combinatorial decision problem. We design a **Risk-Aware Dual-Objective Monte Carlo Tree Search (RA-DUCB-MCTS)** algorithm, which performs tree search under uncertainty to evaluate players' competitive contributions and financial impacts, incorporating injuries and performance fluctuations via stochastic simulations. For the 2025–26 season, the algorithm identifies **Donovan Mitchell** (annual salary $42 million) as the optimal signing, projected to improve team strength by 23.07% with a salary cap utilization of 93%.

To address external structural changes such as league expansion, we construct a **Bi-Level Strategic Equilibrium (BLSE)** system. At the macro level, **heterogeneous mean-field games** simulate inter-team budget competition to solve for the Nash equilibrium, determining the total budget. At the micro level, this budget is fed back as a constraint into the MPC framework to re-optimize roster and marketing strategies. Sensitivity analysis indicates that under direct competition scenarios, the strategy shifts toward *cost efficiency* (e.g., signing **Brandon Ingram**) and reallocates funds to defensive marketing.

The model is further extended to dynamic ticket pricing and injury-adaptive management, enabling data-driven decisions through demand elasticity functions and stochastic recovery models. Simulations across various scenarios verify the framework's strong generalization capability.

This research demonstrates that through systematic modeling, competitive success and financial health can transcend zero-sum game and evolve into a win-win, sustainable growth paradigm.

Keywords: Model Predictive Control (MPC), Monte Carlo Tree Search (MCTS), Bilayer Game, Dynamic Sports Resource Allocation, Competitive-Financial Synergy

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1 Introduction

1.1 Background Information

Contemporary professional sports franchises have evolved from pure championship competitors into sophisticated commercial entities balancing athletic success with profit maximization. As illustrated in Figure 1, under hard salary caps and luxury tax constraints, players function simultaneously as competitive assets and financial investments, forcing management to **reconcile the dual imperatives of winning and fiscal sustainability**.

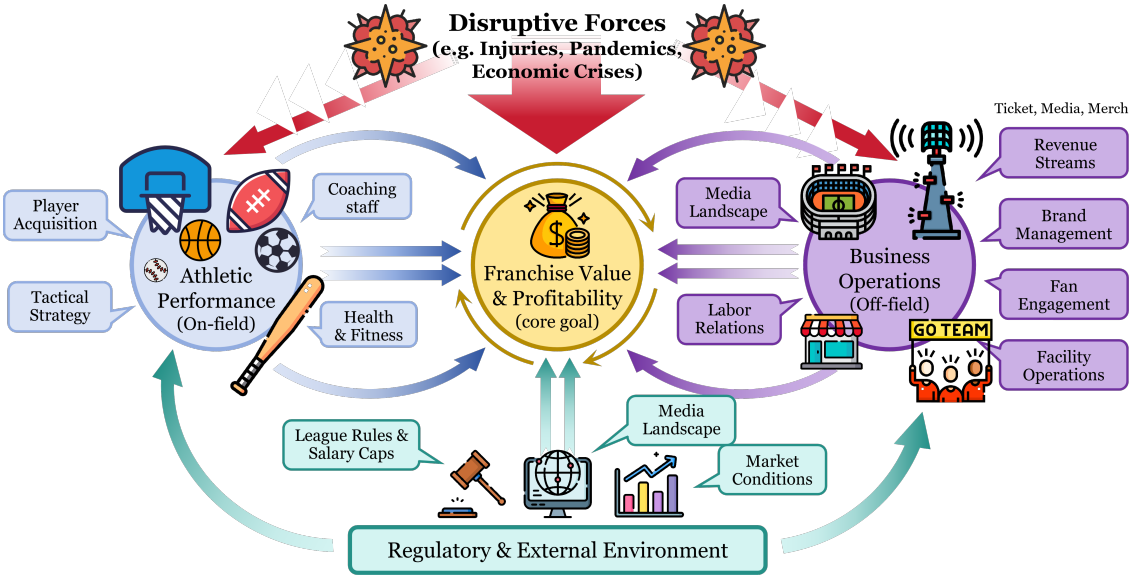


Figure 1: The Integrated System of Modern Sports Franchise Management

While sports analytics standardizes player valuation (e.g., PER, Win Shares), models often neglect commercial impact—some athletes generate substantial revenue despite modest statistics. For commercializing leagues, **integrating analytics with finance to build robust decision systems under uncertainty** critically challenges management.

1.2 Problem Restatement

1. **Task 1: Dynamic Franchise Management.** Develop a control system balancing competitive performance with financial sustainability under evolving economic conditions.
2. **Task 2: Roster Acquisition Strategy.** Optimize player acquisition decisions through free agency under salary cap constraints.
3. **Task 3: League Expansion Impact.** Assess how new team entry affects market dynamics and strategic resource allocation.
4. **Task 4: Dynamic Ticket Pricing.** Determine optimal pricing strategies balancing immediate revenue with long-term fan engagement.
5. **Task 5: Injury-Adaptive Management.** Create decision protocols for managing injured players through retention, release, or rehabilitation pathways.

1.3 Model Overview

The work we have done in this problem is mainly shown in the following 2.

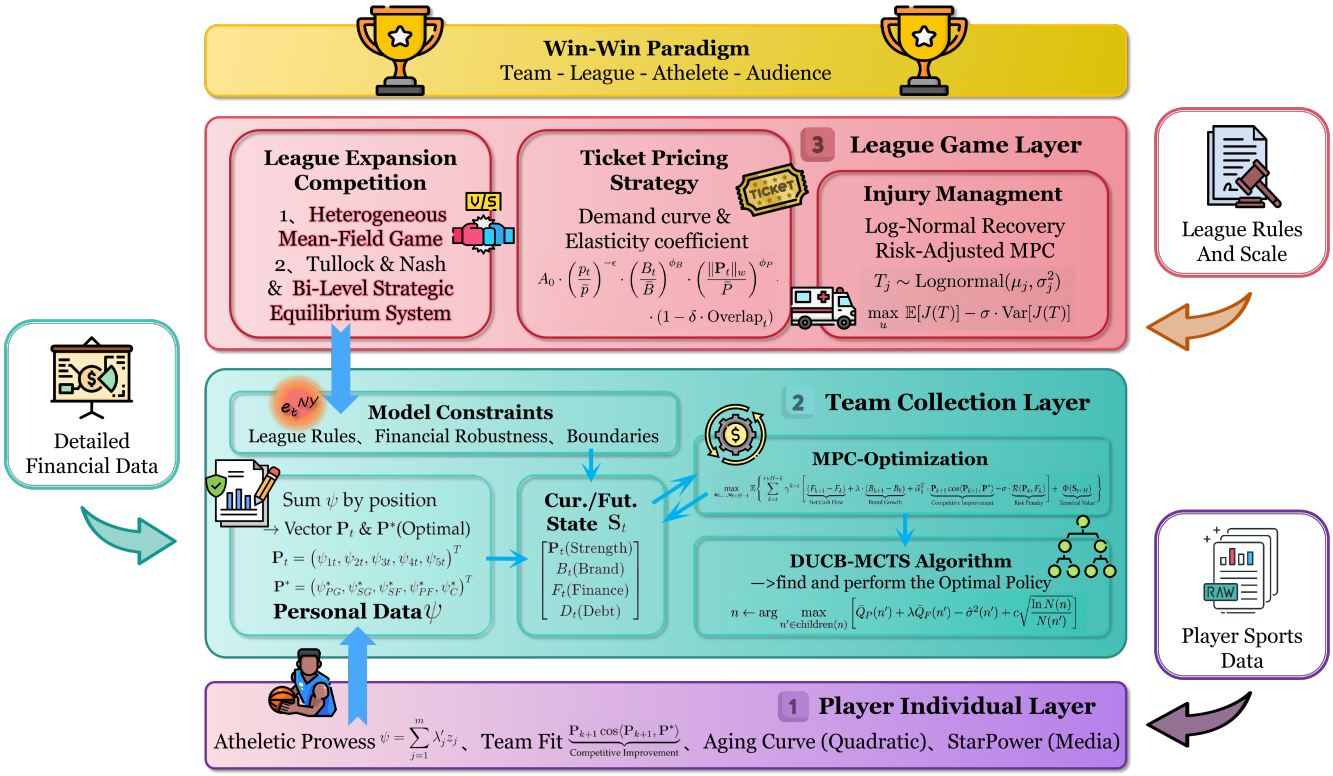


Figure 2: Overall modeling approach and model architecture framework

2 Assumptions

1. **Additive Separability:** Team strength is the weighted sum of player vectors ($\mathbf{P}_{team} = \sum \mathbf{w}_i \cdot \mathbf{P}_i$), enabling independent player valuation.
2. **Rolling-Horizon Optimization:** Decisions are optimized over finite horizons (H years) via MPC, reflecting bounded managerial foresight.
3. **Tullock Contest Equilibrium:** League competition follows $W_i \propto e_i^{\alpha}$ with diminishing returns to spending [1].

3 Notations

Table 1: Key Mathematical Variables

Symbol	Description	Symbol	Description
$\mathbf{P}_t$	Athletic Prowess Vector	B_t	Brand Value Stock
F_t	Financial Reserve Stock	D_t	Debt Stock
$\mathbf{I}_r$	Roster Investment Vector	$I_{\text{marketing}}$	Marketing Investment
p_t	Ticket Price	ψ	Performance Index
$\mathbf{P}^*$	Optimal Athletic Prowess Vector	λ	Brand/Cash Trade-off Weight

4 Task 1: Stochastic Model Predictive Control Framework

4.1 Model Description

As illustrated in Figure 3, our closed-loop MPC system continuously observes franchise state, optimizes multi-period decisions, and updates strategies based on realized outcomes.

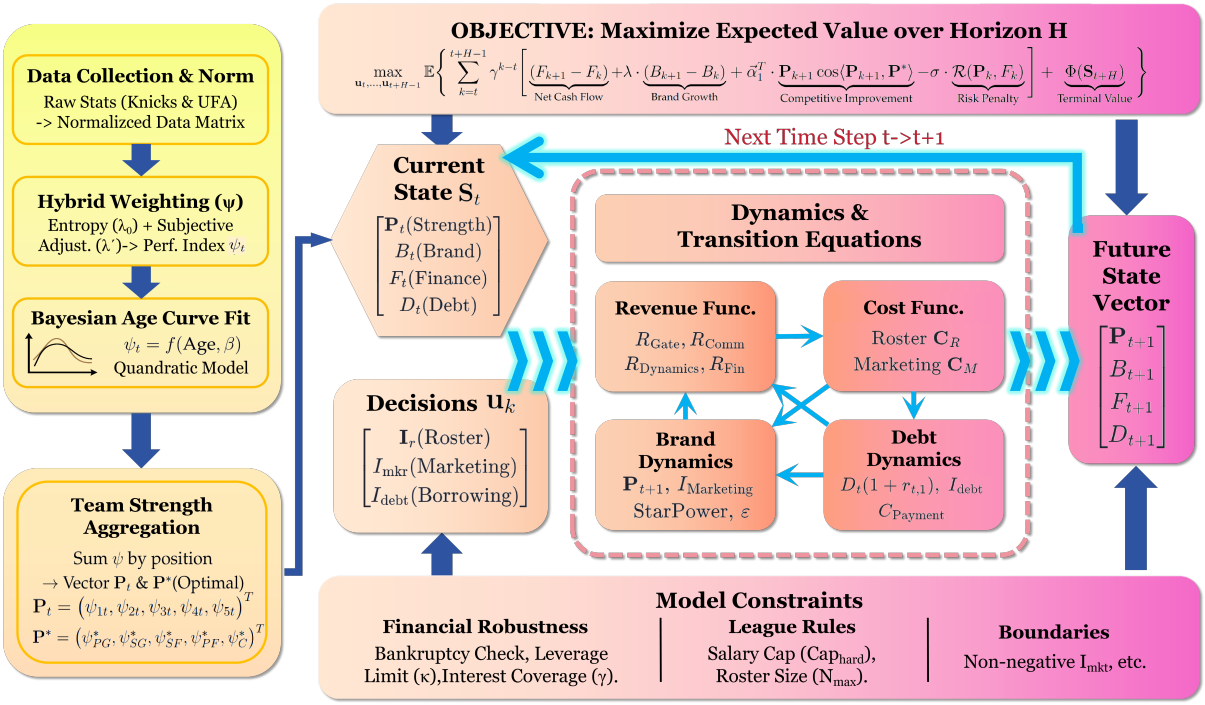


Figure 3: Closed-Loop MPC Control of Franchise Operations

This section develops a comprehensive dynamic decision-making framework, formally designated as the **Model Predictive Control (MPC) Framework with Stochastic Elements**, for professional basketball franchise management. The model integrates three core components:

1. a **Team Performance Metrics** module that quantifies player value through entropy-weighted indicators and aging curve projections;

2. a **State-Space Difference Equations** module that captures the dynamic interplay among athletic assets, brand equity, financial reserves, and debt obligations;

3. a **Optimization Equations** module based on Model Predictive Control (MPC) that balances multi-objective trade-offs under regulatory and financial constraints. Together, these components form a closed-loop system where performance analytics inform financial planning, and strategic decisions feed back into competitive outcomes.

4.2 Team Performance Metrics

Before constructing the state-space dynamics, we establish a framework to quantify individual player athletic value by addressing how to collect and preprocess hetero-

geneous performance data, assign objective and subjective weights to competing metrics, and project future performance trajectories benchmarked against championship standards. The resulting **performance index** ψ and **athletic prowess vector** $\mathbf{P}_t$ serve as key state variables in subsequent sections.

4.2.1 Player Data Collection

We aggregated comprehensive statistics for the **New York Knicks** and potential UFAs, integrating fundamental and advanced metrics (e.g., PER [2], True Shooting %) normalized by minutes played to eliminate playing-time bias. A longitudinal time-series structure tracks each player's career trajectory, identifying critical developmental phases and peak performance windows to **project future asset value within the optimization framework**.

4.2.2 Assigning Subjective and Objective Weightings

We employ the **Entropy Weight Method (EWM)** [3] to derive objective weights $\lambda_{0,j}$ from normalized performance data, ensuring indicators with higher variability dominate the evaluation. To incorporate tactical reality, we apply position-specific subjective corrections via expert-derived scaling factors $\tilde{\lambda}_j$, recognizing that role requirements vary (e.g., rebounds for Centers vs. assists for Guards). The final Performance Index combines both weightings: $\psi = \sum_{j=1}^{m} \lambda'_j z_j$, where $\lambda'_j = \tilde{\lambda}_j / \sum_{l=1}^{m} \tilde{\lambda}_l$ are the normalized composite weights.

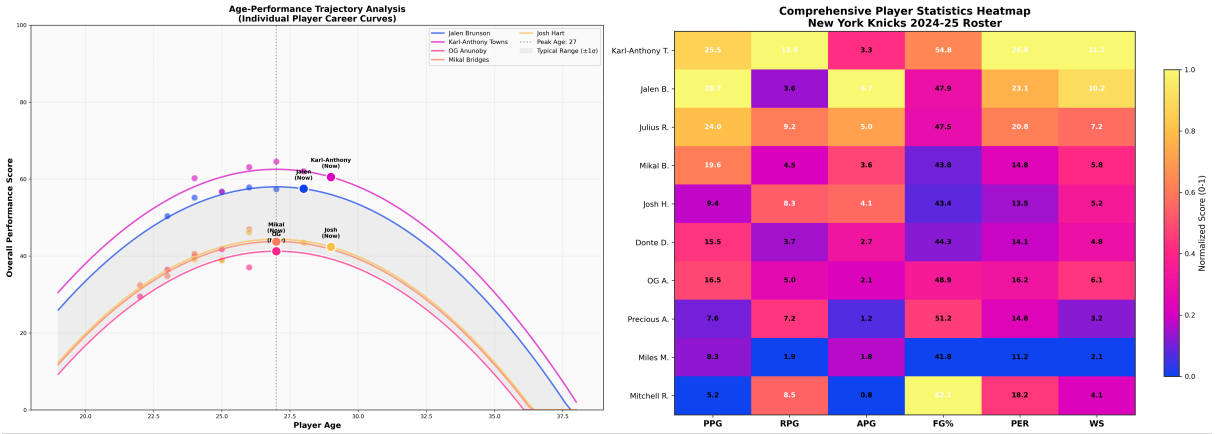


Figure 4: Knicks' Trajectories Age-Performance Curves vs. 2024-25 Stat Heatmap

4.2.3 Longitudinal Performance Trajectory and Aging Curves

To capture temporal evolution of player talent and assess future potential, we construct individualized performance aging curves [4]. We employ a quadratic growth model relating experience level x_t to athletic contribution ψ_t:

$$\psi_t = \beta_0 + \beta_1 x_t + \beta_2 x_t^2 + \epsilon_t \tag{1}$$

where β_1 and β_2 characterize development velocity and curvature, while ϵ_t accounts for stochastic fluctuations. Figure 4 visualizes these aging trajectories for cur-

rent Knicks players, revealing peak performance windows and decline patterns that inform acquisition timing decisions.

4.2.4 Benchmarking via Optimal Athletic Prowess Vectors

Figure 5 presents the salary-performance efficiency frontier, where players above the trend line deliver superior value per dollar while bubble sizes indicate durability through game participation.

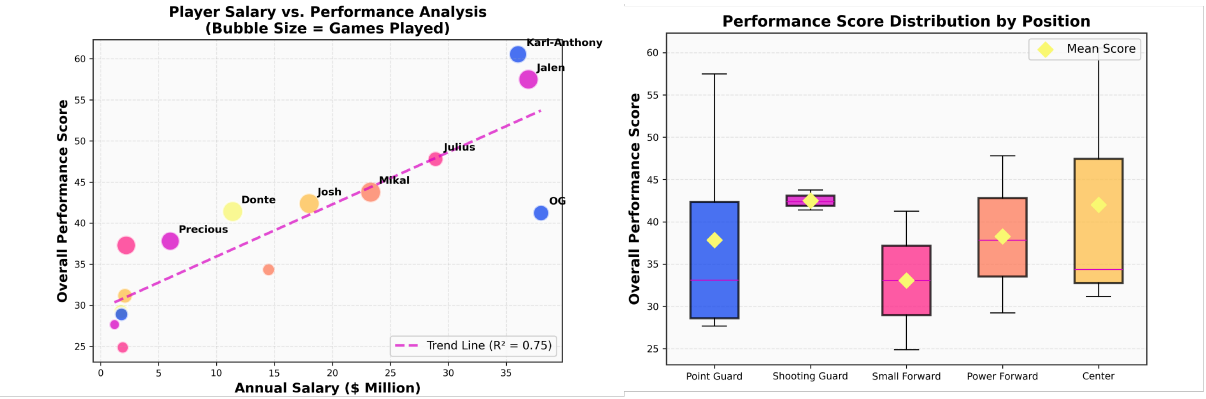


Figure 5: Player Salary vs. Performance Analysis (Bubble Size = Games Played)

To establish a quantitative competitive success target, we analyze historical NBA championship rosters and categorize players into five positions: Point Guard (PG), Shooting Guard (SG), Small Forward (SF), Power Forward (PF), and Center (C). We define the **Optimal Athletic Prowess** vector $\mathbf{P}^*$ by aggregating positional performance indices ψ from championship teams, representing the tactical equilibrium necessary for title contention:

$$\mathbf{P}^* = \left(\psi^*_{PG}, \psi^*_{SG}, \psi^*_{SF}, \psi^*_{PF}, \psi^*_{C}\right)^T, \quad \mathbf{P}_t = \left(\psi_{1t}, \psi_{2t}, \psi_{3t}, \psi_{4t}, \psi_{5t}\right)^T \tag{2}$$

By comparing the Knicks' current vector $\mathbf{P}_t$ against benchmark $\mathbf{P}^*$, management identifies structural deficiencies and prioritizes acquisitions to progressively converge toward championship-caliber configurations, quantifying player value across athletic and commercial dimensions [5]. With the performance quantification framework established, we proceed to embed these metrics within the dynamic state-space system.

The scatter plot reveals the salary-performance efficiency frontier, where players above the trend line (e.g., Brunson, Hart) deliver superior value per dollar, while oversized bubbles indicate durable contributors with high game participation.

4.3 State-Space Difference Equations

To capture the dynamic evolution of a professional basketball franchise across multiple seasons, we formulate the system as a discrete-time state-space model. This framework enables us to track the interplay between athletic performance, brand equity, financial reserves, and debt obligations—while simultaneously optimizing managerial decisions under resource constraints.

4.3.1 State Vector And Decision Vector

We define the complete state representation and management control inputs for season t as:

$$\mathbf{S}_t = \left[\mathbf{P}_t, B_t, F_t, D_t\right]^T, \quad \mathbf{u}_t = \left[\mathbf{I}_r, I_{\text{marketing}}, I_{\text{debt}}\right]^T \tag{3}$$

where:

- $\mathbf{P}_t$ represents the **Athletic Asset Stock** (aggregate competitive capability from Section 4.2)
- B_t quantifies **Brand Value Stock** (intangible assets including fan base, media exposure, reputation)
- F_t reflects **Financial Reserve Stock** (available cash flow and liquid assets)
- D_t tracks **Debt Stock** (cumulative outstanding liabilities)
- $\mathbf{I}_r \in \{0,1\}^{n_{UFA}}$ is the binary roster investment vector where $I_r^j = 1$ signifies signing UFA j at market value
- $I_{\text{marketing}} \geq 0$ allocates capital to commercial operations (ticket pricing, sponsorship)
- $I_{\text{debt}} \geq 0$ represents new debt financing acquired during the current season

4.3.2 System Dynamics

The state transition from season t to $t+1$ is governed by the following system of difference equations:

$$\begin{cases} \mathbf{P}_{t+1} = \displaystyle\sum_{j \in \text{Roster}_t} \mathbf{P}_t^{j,\text{age}+1} + \mathbf{A} \cdot \mathbf{I}_r \\ B_{t+1} = B_t + \vec{\alpha}_1^T(\mathbf{P}_{t+1} - \mathbf{P}_t) + \alpha_2 I_{\text{marketing}} + \text{StarPower}_t + \varepsilon \\ F_{t+1} = F_t + \text{Revenue}(\mathbf{P}_t, B_t) - \text{Cost}(\mathbf{I}_r, I_{\text{marketing}}) - C_{\text{Payment}} \\ D_{t+1} = D_t(1 + r_{t,1}) + I_{\text{debt}} - C_{\text{Payment}} \end{cases} \tag{4}$$

The interpretation of each equation is as follows:

(1) Athletic Asset Dynamics. Asset evolution aggregates the aging-adjusted performance of existing roster players ($\sum \mathbf{P}_t^{j,\text{age}+1}$) and new acquisitions via free agency, where matrix $\mathbf{A}$ maps signing decisions $\mathbf{I}_r$ to positional athletic contributions.

(2) Brand Value Dynamics. Brand equity growth is driven by on-court performance improvements (coefficient $\vec{\alpha}_1$), marketing returns (α_2), and star player appeal (StarPower_t), subject to Gaussian market shocks $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

(3) Financial Reserve Dynamics. This cash flow identity balances liquid assets, incrementing with total revenue while deducting operational costs and debt service payments C_{Payment}.

(4) Debt Dynamics. Liabilities accumulate through accrued interest (rate $r_{t,1}$) and new financing I_{debt}, while being reduced by scheduled repayments C_{Payment}.

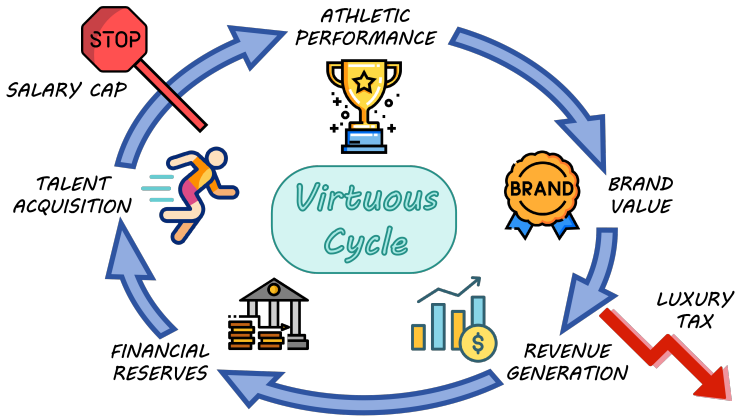


Figure 6: "System Dynamics" Cause-Effect Graph

4.3.3 Cost Function

The total cost function aggregates expenditures from roster management and commercial operations. We model this as a composite function containing base salaries, luxury tax penalties, and operational overhead:

$$\begin{aligned} \text{Cost}(\mathbf{I}_r, I_{\text{marketing}}) &= \mathbf{C}_R(\mathbf{I}_r) + \mathbf{C}_M(I_{\text{marketing}}) \\ \mathbf{C}_R(\mathbf{I}_r) &= \mathbf{w}^T\mathbf{I}_r + \tau \cdot \max\left(0, \mathbf{w}^T\mathbf{I}_r - \text{Cap}_{\text{luxury}}\right) \\ \mathbf{C}_M(I_{\text{marketing}}) &= I_{\text{marketing}} + C_{\text{fixed}} + \phi(\text{Venue}) \end{aligned} \tag{5}$$

The first component, **Athletic Personnel Cost** $\mathbf{C}_R$, accounts for the team's wage bill, where $\mathbf{w}$ is the vector of player salaries. The term involving $\tau > 1$ represents the punitive luxury tax applied when the total payroll exceeds the league-mandated threshold $\text{Cap}_{\text{luxury}}$. The second component, **Marketing and Operational Cost** $\mathbf{C}_M$, sums the direct marketing investment $I_{\text{marketing}}$, fixed administrative overhead C_{fixed}, and venue-specific costs $\phi(\text{Venue})$.

4.3.4 Revenue Function

The total revenue function aggregates four primary income streams: gate receipts, commercial activities, media contracts, and league revenue sharing. We model each component as follows:

$$\begin{aligned} \text{Revenue}_t &= R_{\text{Gate}} + R_{\text{Comm}} + R_{\text{Media}} + R_{\text{Shared}} \\ R_{\text{Gate}} &= C \cdot \text{Att}(\mathbf{P}_t, B_t, I_{\text{mkt}}) \cdot \text{Price}(I_{\text{mkt}}) \\ R_{\text{Comm}} &= \eta_1 B_t + \eta_2 \cdot \text{StarPower}(\mathbf{P}_t) \\ R_{\text{Media}} &= R_{\text{base}} + \theta \cdot \text{Rank}(\mathbf{P}_t) \\ R_{\text{Shared}} &= \gamma \cdot (\bar{B}_{\text{League}} - B_t) \end{aligned} \tag{6}$$

The revenue framework encompasses four integrated streams. **Direct Operations** include gate revenue R_{Gate} (venue capacity × attendance × pricing) and commercial revenue R_{Comm} (merchandising and sponsorship driven by brand strength $\eta_1 B_t$ and player marketability $\eta_2 \cdot \text{StarPower}$), both directly influenced by team performance and marketing strategies.

Level Distributions comprise media revenue R_{Media} (guaranteed broadcasts R_{base} plus performance bonuses $\theta \cdot \text{Rank}$) and revenue sharing R_{Shared} (redistribution transfers where $\gamma(\bar{B}_{\text{League}} - B_t)$ provides subsidies to below-average teams), establishing the collective economic foundation independent of individual franchise control.

Having specified the complete state-space dynamics, we now formulate the optimization problem that determines optimal decision trajectories.

4.4 Construction of Optimization Equations

We formulate management decisions as Model Predictive Control (MPC) [6], synthesizing sports analytics with finance. This rolling-horizon approach offers three advantages:

- **Multi-Objective Balancing:** MPC balances competing goals—immediate solvency vs. long-term championships—in one function.
- **Adaptive Re-optimization:** The receding horizon continuously re-optimizes as new information arrives (injuries, performances), mirroring NBA dynamics.
- **Regulatory Compliance:** MPC embeds league rules (caps, roster sizes) as constraints, ensuring policies comply with regulations.

4.4.1 Objective Function Formulation

The franchise's management seeks to maximize the expected discounted utility over a finite planning horizon H, balancing immediate cash flows against long-term asset appreciation while penalizing excessive risk exposure. The objective function is formulated as:

$$\max_{\mathbf{u}_t,\ldots,\mathbf{u}_{t+H-1}} \mathbb{E}\Bigg\{ \sum_{k=t}^{t+H-1} \gamma^{k-t} \Bigg[\underbrace{(F_{k+1} - F_k)}_{\text{Net Cash Flow}} + \lambda \cdot \underbrace{(B_{k+1} - B_k)}_{\text{Brand Growth}} + \vec{\alpha}_1^T \cdot \underbrace{\mathbf{P}_{k+1} \cos\langle \mathbf{P}_{k+1}, \mathbf{P}^* \rangle}_{\text{Competitive Improvement}} - \sigma \cdot \underbrace{\mathcal{R}(\mathbf{P}_k, F_k)}_{\text{Risk Penalty}} \Bigg] + \underbrace{\Phi(\mathbf{S}_{t+H})}_{\text{Terminal Value}} \Bigg\} \tag{7}$$

The components of this objective function reflect the multi-dimensional nature of franchise value creation:

(1) Net Cash Flow $(F_{k+1} - F_k)$ captures changes in liquid reserves, ensuring operational sustainability and investment capacity.

(2) Brand Value Appreciation $\lambda(B_{k+1} - B_k)$ weights intangible asset growth relative to immediate cash returns, where $\lambda > 0$ reflects management's brand-building priority.

(3) Competitive Strength Improvement. The term $\mathbf{P}_{k+1} \cos\langle \mathbf{P}_{k+1}, \mathbf{P}^* \rangle$ measures both the magnitude and directional alignment of the team's athletic prowess vector with the championship benchmark $\mathbf{P}^*$. The cosine similarity ensures that roster construction not only increases aggregate talent but also optimizes positional balance toward proven winning configurations.

(4) Terminal Value Function. To prevent myopic decision-making, we define the going-concern value as $\Phi(\mathbf{S}_{t+H}) = \omega_1 F_{t+H} + \omega_2 B_{t+H}$, where capitalization weights

ω_1, ω_2 reflect the perpetuity value of financial reserves and brand equity.

4.4.2 Constraint System

The optimization is subject to four categories of constraints that reflect physical, financial, and regulatory realities. For all $k \in \{t, t+1, \ldots, t+H-1\}$, we impose:

$$\begin{cases} \mathbf{S}_{k+1} = f(\mathbf{S}_k, \mathbf{u}_k) & D_k \leq \kappa \cdot B_k \\ F_{k+1} = F_k + \Delta F_k \geq C_{\text{fixed}} & \dfrac{\text{Revenue}_k}{r_{k,1} \cdot D_k} \geq \xi \\ I_{\text{roster},k} + I_{\text{marketing},k} + r_k D_k \leq e_k^{NY} & \mathbf{w}^T \cdot \mathbf{I}_r \leq \text{Cap}_{\text{hard}} \\ \mathbf{I}_r \in \{0,1\}^{n_{UFA}},\ I_{\text{marketing}} \geq 0,\ I_{\text{debt}} \geq 0 & \|\mathbf{I}_r\|_0 \leq N_{\max} \end{cases} \tag{8}$$

(1) State Transition Constraints. The difference equations established in Section 5.3.3 ensure that all projected trajectories are physically realizable given the system dynamics.

(2) Financial Stability Constraints. Three prudential requirements maintain franchise viability: the financial reserve must remain sufficient to cover fixed operational costs (bankruptcy protection), total debt cannot exceed a fraction κ of brand equity (leverage limits), and operating revenues must cover debt service obligations by a factor of ξ (interest coverage).

(3) League Regulatory Constraints. Professional sports leagues require roster expenditure to respect the salary hard cap and roster size to not exceed the maximum allowable number of signed players.

(4) Decision Variable Bounds. All control inputs must satisfy non-negativity and integrality requirements.

5 Task 2: Policy Solver Based on Monte Carlo Tree Search

The MPC framework presents a combinatorial optimization challenge: selecting from $2^{n_{UFA}}$ possible roster configurations under mixed-integer constraints. Traditional gradient-based methods fail due to the discrete decision space and non-convex objective landscape. We address this by developing a novel **Risk-Aware Dual-UCB Monte Carlo Tree Search (RA-DUCB-MCTS)** algorithm [7] that integrates financial and athletic objectives while handling uncertainty through stochastic rollouts.

5.1 Problem Reformulation as Sequential Decision Game

The RA-DUCB-MCTS algorithm is designed to solve the complex combinatorial optimization problem of roster acquisition under salary cap constraints. The algorithm balances dual objectives—athletic performance enhancement and financial return maximization—while managing uncertainty through risk-aware tree search and stochastic rollouts.

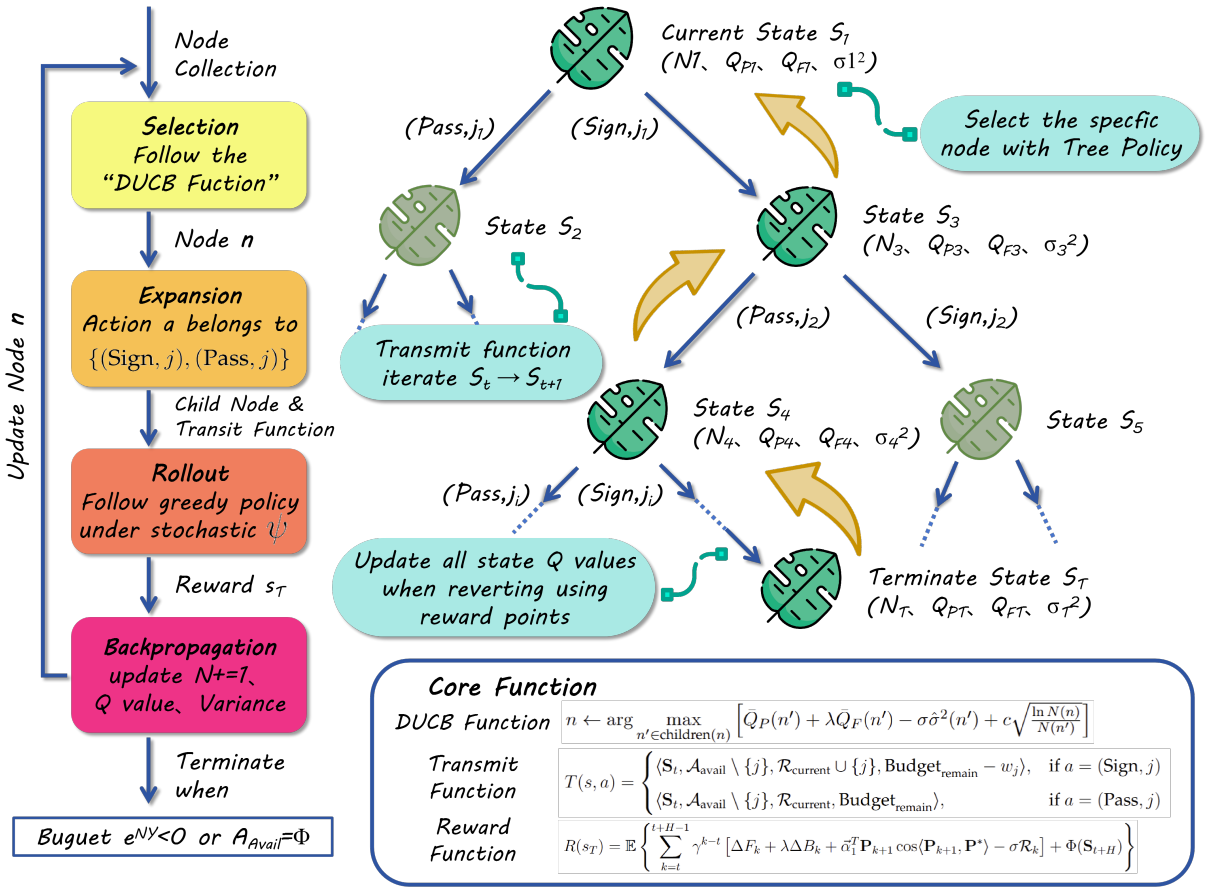


Figure 7: Risk-Aware Dual-UCB Monte Carlo Tree Search Algorithm Flowchart

5.2 MCTS-MPC Integration Algorithm

Algorithm 1 RA-DUCB-MCTS Policy Solver for MPC Roster Optimization

Input: MPC state $\mathbf{S}_t$, UFA pool $\mathcal{U}$, Hard cap Cap_{hard}, Horizon H, Iterations N_{iter}, Weights (λ, σ)
Output: Optimal signing decisions $\mathbf{I}_r^*$ for current MPC epoch

Initialize root n_0 from MPC state $\mathbf{S}_t$; prune infeasible UFAs ▷ Sync state & enforce constraints
for $i = 1$ to N_{iter} **do**
  $n \leftarrow n_0$
  while n is fully expanded and non-terminal **do** ▷ Selection via dual objectives
    $n \leftarrow \arg\max_{n' \in \text{children}(n)} \left[\bar{Q}_P(n') + \lambda \bar{Q}_F(n') - \sigma\hat{\sigma}^2(n') + c\sqrt{\frac{\ln N(n)}{N(n')}}\right]$
  end while
  if n is non-terminal **then** ▷ Expansion with cap compliance
    Expand next UFA: create child for (Sign, j) or (Pass, j) if feasible
  end if
  Rollout from $s \leftarrow \text{State}(n)$ over horizon H with stochastic ψ; stop at s_T
  Evaluate $R(s_T)$ via MPC objective (Eq. 26) and backpropagate to root
end for
return $\mathbf{I}_r^*$ from most-visited child of n_0

5.3 Result Analysis

5.3.1 Evaluation Methodology Recap

Building upon the Team Performance Metrics established in Section 4.2, our player evaluation framework synthesizes seven weighted dimensions: *Production* (30%), *Efficiency* (20%), *Team Fit* (15%), *Age Curve* (10%), *Shooting* (10%), *Cost Efficiency* (8%), and *Durability* (7%). This multi-factor scoring function transforms heterogeneous player attributes into a unified **Comprehensive Value Score**, enabling direct comparison across positions. For team dynamics, we measure roster composition against the championship benchmark $\mathbf{P}^*$ via cosine similarity, identifying the Knicks' critical deficiency at the Shooting Guard position — a gap that drives our acquisition priority.

5.3.2 Optimal Strategy Output

Executing the RA-DUCB-MCTS algorithm with $N_{\text{iter}} = 10,000$ iterations yields the following decision vector $\mathbf{u}^*$ for the 2025-26 season:

Table 2: MPC-MCTS Optimization Results for 25-26 Season

Decision Variable	Optimal Value	Constraint Status
Roster Investment $\mathbf{I}_r^*$	Sign **Donovan Mitchell**	Within Cap
Player Salary	\$42M / year	93% of Available Cap
Marketing Investment $I^*_{\text{marketing}}$	$\approx$ \$16M	Aligned with Brand Growth
Debt Financing I^*_{debt}	$\approx$ \$0.5M	$< 1\%$ of Brand Equity

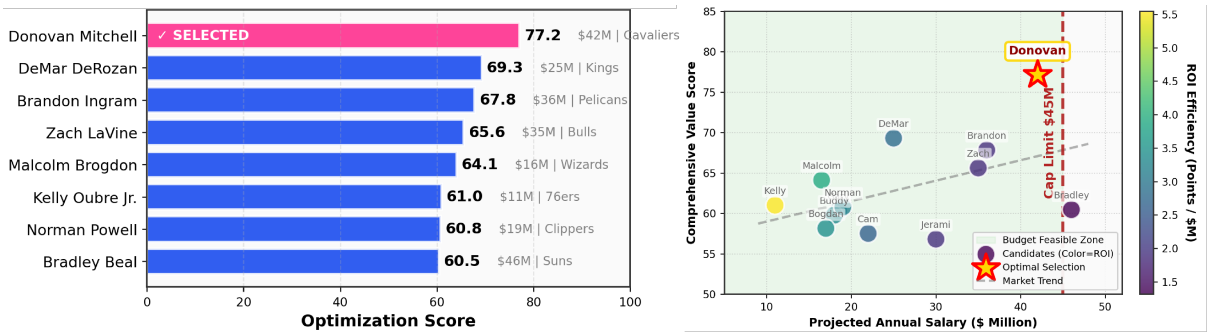


Figure 8: Acquisition Optimization: Value-Cost Landscape and Selection Results

The algorithm identifies **Donovan Mitchell** as the singular optimal acquisition, achieving the highest composite score (78.2) among all UFA candidates. As illustrated in Figure 8, Mitchell dominates the value-cost efficiency frontier — his position as the gold star above the market trend line indicates superior return on investment relative to salary expectations.

5.3.3 Strategic Advantages and Risk Assessment

The recommended strategy offers compelling **commercial advantages** by maximizing both competitive and financial utility. (1) **Positional Gap Closure**: Mitchell addresses the critical Shooting Guard deficiency, directly optimizing the vector alignment $\mathbf{P}_{t+1} \cos\langle \mathbf{P}_{t+1}, \mathbf{P}^* \rangle$. (2) **Star Power Synergy**: His marketability amplifies brand

equity (StarPower$_t$), while the \$16M marketing allocation leverages his signing to boost revenue streams. (3) **Financial Efficiency**: The accommodative monetary environment minimizes the cost of the \$0.5M debt financing, preserving liquidity.

However, the strategy entails calculated **risks**: (1) **Concentration Risk**: Allocating 93% of cap space to a single asset reduces flexibility against injury or underperformance. (2) **Delayed ROI**: The \$42M commitment prioritizes long-term brand appreciation over immediate cash flow. (3) **Macro-Sensitivity**: Potential interest rate hikes or negotiation premiums could compress margins.

In summary, the MPC-MCTS framework recommends an *aggressive but calculated* strategy, prioritizing championship-driven brand growth over short-term conservatism.

6 Task 3: Bi-Level Strategic Equilibrium System

The MPC framework developed in Sections 4–5 optimizes single-team operations under a *fixed* league environment. However, when the NBA expands by adding new franchises, the competitive landscape fundamentally changes. To address this, we construct a consolidated **Bi-Level Strategic Equilibrium (BLSE) System** that captures league-wide strategic interactions and their propagation to team-level decisions.

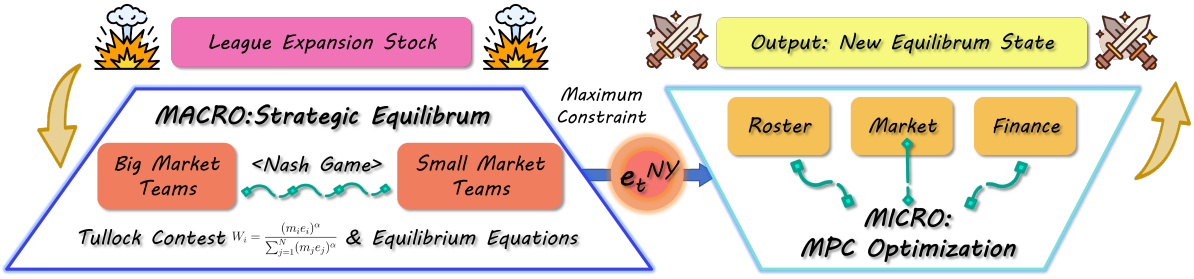


Figure 9: Bi-Level Strategic Equilibrium System Framework

6.1 Bi-Level Model Architecture

We adopt a hierarchical structure separating "*how much to spend*" (macro equilibrium) from "*how to spend it*" (micro optimization):

- **Macro Layer (Static Game):** All teams simultaneously choose total expenditure e_i in a one-shot Nash game, determining league-wide competitive balance.
- **Micro Layer (Dynamic Optimization):** Given the equilibrium expenditure e^*_{NY} from above, the Knicks solve the MPC problem (Section 5.4) with an additional budget constraint.

This architecture reflects real-world decision-making: ownership first sets seasonal budgets based on competitive positioning, then management allocates resources optimally within that envelope.

6.2 Macro Layer: Heterogeneous Mean-Field Game

We partition all N teams into big-market ($\mathcal{B}$) and small-market ($\mathcal{S}$) categories based on metropolitan population, with the New York Knicks belonging to $\mathcal{B}$ at normalized market size $m_{NY} = 1.0$. Each team's competitive success is modeled via a **Tullock**

contest function, formulating inter-team competition as a Nash game [8] within a dynamic framework [9]:

$$W_i = \frac{(m_i e_i)^\alpha}{\sum_{j=1}^{N}(m_j e_j)^\alpha}, \quad \text{where } \alpha \in (0,2) \tag{9}$$

This captures the intuition that larger markets achieve greater impact per dollar spent, while maintaining the zero-sum constraint $\sum_i W_i = 1$(Typically, we set α equal to 1 here). Each team maximizes single-period profit:

$$\pi_i = [R_0 + R_1 W_i] m_i - \frac{k}{2} e_i^2$$

balancing revenue gains from winning against quadratic costs reflecting diminishing returns to spending. Applying first-order conditions and imposing symmetric strategies within each class yields the Nash equilibrium system:

$$\begin{cases} R_1 m_B \cdot \dfrac{\alpha W_B(1-W_B)}{e_B} = k e_B \\ R_1 m_S \cdot \dfrac{\alpha W_S(1-W_S)}{e_S} = k e_S \end{cases} \quad \text{where : } \begin{cases} W_B = \dfrac{N_B(m_B e_B)^\alpha}{N_B(m_B e_B)^\alpha + N_S(m_S e_S)^\alpha} \\ W_S = \dfrac{N_S(m_S e_S)^\alpha}{N_B(m_B e_B)^\alpha + N_S(m_S e_S)^\alpha} \end{cases} \tag{10}$$

Solving yields equilibrium expenditures (e_B^*, e_S^*), with the Knicks' optimal budget $e_{NY}^* = e_B^*$.

The equilibrium derived above characterizes a *static* league structure with fixed N teams. However, when the NBA expands by adding new franchises, this equilibrium is fundamentally perturbed. Unlike routine seasonal fluctuations, expansion represents a *structural shock* that simultaneously affects competitive balance, revenue distribution, and market segmentation. Our modeling approach captures this disruption through two key mechanisms: **market cannibalization** (direct fan diversion) and **league growth** (expanded total market size).

6.2.1 New Franchise Characteristics

Each expansion team is characterized by two critical parameters that jointly determine its impact on incumbents:

- **Market size** $m_{new} \in (0,1]$: Normalized against the largest existing market (New York = 1.0), this captures the economic potential of the new franchise's metropolitan area—including population, median income, and sports consumption propensity.
- **Geographic overlap** $\text{Overlap}(i, \text{new}) = 1/(1+d_i)$: This function quantifies the degree of fan-base competition between the new team and existing team i, where d_i represents normalized geographic distance. Teams in the same city ($d_i \approx 0$) approach Overlap ≈ 1, while distant markets ($d_i \gg 1$) exhibit negligible overlap.

The interaction between these parameters creates distinct expansion archetypes: a high-m_{new}, high-overlap scenario represents direct competitive threat (e.g., a second New York team), while high-m_{new}, low-overlap represents market expansion opportunity (e.g., Seattle entry).

6.2.2 Effective Market Adjustment

Post-expansion, incumbent teams experience market dilution proportional to their geographic proximity to the new entrant, while simultaneously benefiting from league-wide market expansion. We model this dual effect through a two-stage adjustment process:

$$\begin{aligned} \text{Stage 1 (Diversion):} \quad & m_i' = m_i \cdot \prod_{j \in \text{New}} (1 - \beta \cdot \text{Overlap}(i,j)) \\ \text{Stage 2 (Reallocation):} \quad & \tilde{m}_i = (1+\theta) \cdot M_{\text{total}} \cdot \frac{m_i'}{\sum_{k=1}^{N+n} m_k'} \end{aligned} \tag{11}$$

where $\beta = 0.3$ is the *diversion coefficient* calibrated from historical NBA expansion episodes (Charlotte 2004, Memphis 2001), and $\theta \in [0.05, 0.15]$ represents the *league growth factor*.

This mechanism balances **local cannibalization** (Stage 1: fan diversion to nearby entrants) against **global growth** (Stage 2: total market expansion by $1+\theta$). The net impact on $\tilde{m}_i$ depends on whether the systemic "growth dividend" offsets localized losses, typically favoring distant teams while penalizing direct geographic rivals.

6.2.3 Re-Equilibration Process

With updated market parameters $\{m_i'\}$ and expanded league size $N' = N + n$, we re-solve the Nash equilibrium system (Eq. 33) to obtain new optimal expenditures $(e_B^{*'}, e_S^{*'})$. The comparative statics yield the Knicks' budget adjustment:

$$\Delta e_{NY} = \frac{e_{NY}^{*'} - e_{NY}^{*}}{e_{NY}^{*}} \tag{12}$$

This percentage change encapsulates how the altered competitive landscape—more teams competing for wins, potentially diluted market share, but also expanded league revenues—translates into revised optimal spending levels. Notably, Δe_{NY} can be positive or negative depending on expansion location, reflecting the non-trivial strategic implications of league structure changes.

6.3 Micro Layer: Strategy Adjustment

The bi-level architecture ensures **minimal modification** to our established MPC framework while incorporating league-level dynamics. Rather than rebuilding the optimization from scratch, we inject the macro equilibrium result as a single additional constraint—embodying the principle that ownership sets the budget envelope, while management optimizes allocation within it.

The modified constraint system adds:

$$\text{Budget Constraint:} \quad \mathbf{w}^T \mathbf{I}_r + I_{\text{marketing}} + r_t D_t \leq e_{NY}^{*'} \tag{13}$$

This formulation preserves all original state dynamics, objective function components (Eq. 26), and regulatory constraints (Eq. 28). The MPC-MCTS solver (Algorithm

1) operates identically, but now explores a potentially contracted or expanded feasible region depending on $e^{*'}_{NY}$.

The elegance of this approach lies in its *modularity*: the macro game determines "how much to spend," while the micro optimization determines "how to spend it." This separation mirrors actual organizational decision-making, where board-level strategic budgeting precedes departmental operational planning.

6.4 Location Impact Analysis

To operationalize the expansion model, we analyze three canonical scenarios representing the spectrum of possible expansion configurations. Each scenario is evaluated by computing the equilibrium shift and tracing its propagation through the MPC framework.

Table 3: Expansion Location Impact on Knicks' Optimal Budget

Scenario	m_{new}	Overlap(NY)	**Knicks Impact**
S1: Direct Competition (Brooklyn II)	0.9	0.8	$\Delta e_{NY} \approx -18\%$
S2: Blue Ocean (Seattle)	0.9	0.1	$\Delta e_{NY} \approx +3\%$
S3: Small Market (Louisville)	0.3	0.2	$\Delta e_{NY} \approx -4\%$

We highlight three distinct archetypes: **S1 (Direct Competition)** triggers an 18% "defensive contraction" due to severe cannibalization (0.8 overlap); **S2 (Blue Ocean)** enables a 3% budget rise via global revenue growth; and **S3 (Small Market)** remains neutral. Synthesizing these patterns, expansion structurally **harms** incumbents facing high geographic overlap or financial fragility, while **benefiting** those situated to harvest league-wide growth dividends without local rivalry.

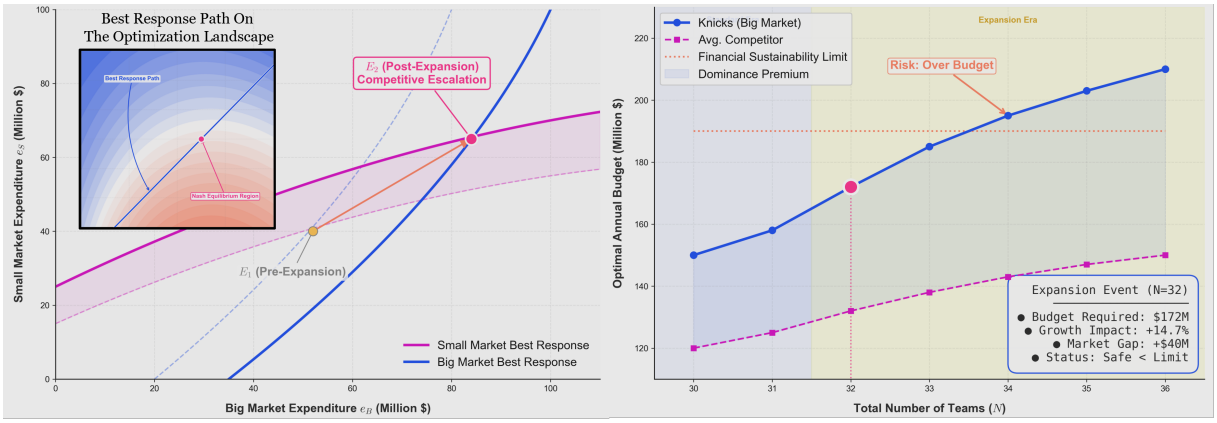


Figure 10: Nash Equilibrium Shift, Profit Landscape, and Sensitivity Analysis

Figure 10 visualizes the expansion mechanism through three perspectives. The left panel demonstrates league expansion as an external shock, shifting best response curves outward and driving equilibrium from E_1 to E_2—a "competitive arms race" effect. The upper left panel maps the profit optimization landscape, where the Nash equilibrium (pink point) sits on the ridge line representing optimal expenditure; deviations into darker regions yield suboptimal returns. The right panel quantifies impacts

through sensitivity analysis: under $N = 32$ expansion, the Knicks' budget surges 12% above average competitors (blue "Dominance Premium" region). Critically, the orange dashed line marks the **Financial Sustainability Limit**—expansion beyond $N = 34$ breaches this threshold, rendering on-court success financially unsustainable.

6.5 Strategy Adjustment Recommendations

Table 4: Task 3 vs. Baseline: Strategic Adjustment Summary

Metric	Baseline	Expansion (S1)	Change
Budget Constraint	Unconstrained	$\downarrow 18\%$	Defensive
Acquisition Target	D. Mitchell ($42M)	B. Ingram ($33M)	Cost-efficient
Performance Gain	+23%	+21%	-2%
Marketing Focus	Growth	Defense	$+\$4M$
Financial Leverage	Conservative	Moderate	Strategic

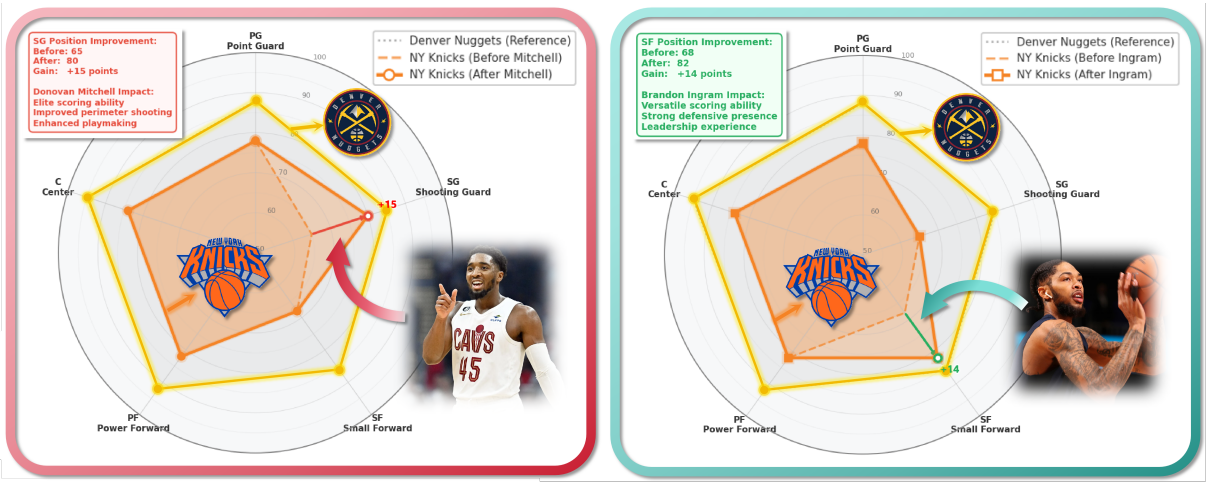


Figure 11: New York Knicks adjust signing strategy

The bi-level optimization framework yields concrete, scenario-contingent guidance, as visualized in Figure 11. While the base model (Task 1&2) recommends signing **Donovan Mitchell** (23.07% skill enhancement) to maximize pure competitive strength, the direct competition scenario (Task 3, S1) drastically alters the optimal strategy. Under this pressure, the framework pivots to **Brandon Ingram** (21.21% improvement), whose lower market value ($33M vs. $42M) epitomizes a shift toward *cost-efficient acquisition*. This strategic reallocation frees capital for defensive marketing ($16M $\rightarrow$ $20M) to protect market share, supported by slightly elevated debt financing ($0.5M $\rightarrow$ $0.8M) that leverages accommodative monetary policy.

In contrast, **S2 (Blue Ocean)** enables **aggressive expansion**: channeling budget surplus into premium roster upgrades and national branding to seize the championship window, while **S3 (Small Market)** calls for **efficiency optimization**: maintaining baseline strategies while refining cost-per-win metrics without structural upheaval. In all scenarios, the receding-horizon nature of MPC ensures adaptive response capability, progressively refining strategy toward the evolving equilibrium.

7 Task 4 & 5: Pricing Strategy and Injury Management

The MPC framework established in Sections 4–5 operates under implicit assumptions: fixed ticket pricing and a healthy roster. Real-world franchise management, however, must dynamically adjust pricing to maximize gate revenue and adapt operations when key players suffer injuries. This section extends the MPC formulation to address these two interconnected challenges within a unified optimization structure.

7.1 Task 4: Dynamic Ticket Pricing Strategy

Ticket sales exhibit significant variance driven by venue capacity, seasonality, team popularity, opponent strength, and market size. Management faces a fundamental trade-off: maximize per-game revenue through premium pricing, or stimulate attendance to cultivate long-term fan loyalty and season-ticket conversion. We embed this decision into the MPC framework by introducing ticket price as an explicit control variable.

7.1.1 Decision Variable Extension

The control vector is augmented to include seasonal average ticket price:

$$\mathbf{u}_t = [\mathbf{I}_{r,t}, I_{\text{marketing},t}, I_{\text{debt},t}, p_t]^T \tag{14}$$

where $p_t \in \mathbb{R}^+$ represents the *strategic average ticket price* for season t. This aggregate formulation captures board-level pricing policy, while game-by-game adjustments (opponent premiums, promotional discounts) are treated as tactical variations around this baseline.

7.1.2 State-Dependent Demand Function

Attendance demand is modeled as a **constant-elasticity** function of price, brand equity, and competitive strength, employing demand elasticity modeling [10, 11]:

$$A_t = \min\left\{ A_0 \cdot \left(\frac{p_t}{\bar{p}}\right)^{-\epsilon} \cdot \left(\frac{B_t}{\bar{B}}\right)^{\phi_B} \cdot \left(\frac{\|\mathbf{P}_t\|_w}{\bar{P}}\right)^{\phi_P} \cdot (1 - \delta \cdot \text{Overlap}_t),\ C \right\} \tag{15}$$

where $A_t \in [0, 1]$ denotes expected attendance rate, $\epsilon > 0$ is the price elasticity, $\phi_B, \phi_P > 0$ are brand and performance elasticities respectively, $\|\mathbf{P}_t\|_w = \sum_i w_i \psi_i$ is the weighted roster strength, Overlap_t captures market competition from expansion (Section 6), and C is venue capacity.

The power-law specification ensures that elasticity coefficients directly measure percentage responsiveness: a 1% price increase reduces demand by ϵ%, while equivalent improvements in brand or performance boost attendance by ϕ_B% and ϕ_P% respectively. This log-linear structure facilitates empirical calibration via regression on historical ticket data.

7.1.3 Revenue Function Update

The gate revenue component in the MPC objective is updated as:

$$R_{\text{Gate}}(p_t, B_t, \mathbf{P}_t) = N_{\text{games}} \cdot C \cdot A_t \cdot p_t \tag{16}$$

This formulation creates an *endogenous feedback loop*: higher prices directly increase per-ticket revenue but suppress attendance through the elasticity mechanism, while investments in roster strength ($\mathbf{I}_r$) and marketing ($I_{\text{marketing}}$) enhance both $\mathbf{P}_t$ and B_t, shifting the demand curve outward.

7.1.4 Optimal Pricing Logic

The MPC framework automatically resolves the short-term vs. long-term trade-off:

- **Rebuilding Phase** ($\|\mathbf{P}_t\|_w$ low): Optimal p_t^* is suppressed to maintain attendance, nurturing fan loyalty that compounds into future brand value B_{t+k}.
- **Contending Phase** ($\|\mathbf{P}_t\|_w$ high): Optimal p_t^* rises to capture championship-driven demand surge, monetizing competitive success.
- **Expansion Shock** ($\text{Overlap}_t > 0$): Market competition systematically depresses p_t^* to defend market share against new entrants.

Table 5 presents key ticket pricing parameters and results for each team phase. The elasticities ϵ, ϕ_B, and ϕ_P show how price, brand, and performance affect attendance. As teams move from rebuilding to championship contention, both optimal price (p_t^*) and attendance (A_t^*) rise, reflecting stronger demand. After league expansion, both values drop, highlighting the impact of increased competition. The table illustrates the need to adjust pricing as team status and market conditions change.

Table 5: Demand Elasticity Parameters and Optimal Pricing by Team Phase

Team Phase	ϵ	ϕ_B	ϕ_P	p_t^*	A_t^*
Rebuilding (Low $\|\mathbf{P}_t\|_w$)	0.40	0.20	0.25	\$85	92.3%
Developing (Mid $\|\mathbf{P}_t\|_w$)	0.35	0.25	0.30	\$115	88.7%
Contending (Knicks 2024-25)	**0.35**	**0.25**	**0.30**	**\$142**	**91.5%**
Championship Window	0.30	0.30	0.35	\$185	95.2%
Post-Expansion (S1: High Overlap)	0.38	0.22	0.28	\$118	85.4%

Note: ϵ = price elasticity, ϕ_B = brand elasticity, ϕ_P = performance elasticity.
Optimal values derived from MPC framework maximizing $R_{\text{Gate}} = N_{\text{games}} \cdot C \cdot A_t \cdot p_t$.

7.2 Task 5: Injury-Adaptive Management

Player injuries represent exogenous shocks that simultaneously degrade competitive strength and create complex financial trade-offs: retain injured players (incurring salary costs without performance), release them (triggering contract buyouts), or invest in accelerated rehabilitation. We extend the MPC framework to optimize these decisions under uncertainty about recovery timelines, handling stochastic recovery times and risk-adjusted decisions [12].

7.2.1 Extended Decision Variables

Let $\mathcal{J}_t \subseteq \text{Roster}_t$ denote the set of injured players at season t. We introduce:

$$\mathbf{u}_t = [\mathbf{I}_{r,t}, I_{\text{marketing},t}, I_{\text{debt},t}, p_t, I_{\text{rehab},t}, \mathbf{v}_t]^T \tag{17}$$

where $I_{\text{rehab},t} \geq 0$ is rehabilitation investment accelerating recovery, and $\mathbf{v}_t \in \{0,1\}^{|\mathcal{J}_t|}$ is the retention decision vector ($v_j = 1$: retain player j; $v_j = 0$: release with buyout).

7.2.2 Modified State Transition Dynamics

The injury shock propagates through a coupled state transition system:

$$\begin{aligned}
\mathbf{P}_{t+1} &= \sum_{j \in \text{Roster}_t \setminus \mathcal{J}_t} \mathbf{P}_j^{(\text{age}+1)} + \mathbf{A} \cdot \mathbf{I}_{r,t} \\
\text{StarPower}_{t+1} &= \text{StarPower}_t - \sum_{j \in \mathcal{J}_t} v_j \cdot \text{Influence}_j \\
F_{t+1} &= F_t + \text{Revenue}_t - \text{Cost}_t - I_{\text{rehab},t} - \sum_{j \in \mathcal{J}_t} (1 - v_j) \cdot \text{Buyout}_j
\end{aligned} \tag{18}$$

In this system, injuries reduce effective on-court strength by excluding $\mathcal{J}_t$ from the roster aggregation, while star unavailability depresses marketability through an influence-weighted loss in StarPower. Financially, management chooses whether to retain each injured player ($v_j = 1$) or release them ($v_j = 0$), trading ongoing roster continuity against buyout payments; rehabilitation spending $I_{\text{rehab},t}$ further shifts this trade-off by accelerating recovery.

7.2.3 Recovery Modeling and Risk-Adjusted Objective

Recovery time T_j for injured player j follows a **log-normal distribution** [13]: $T_j \sim \text{Lognormal}(\mu_j, \sigma_j^2)$ with $\mathbb{E}[T_j] = e^{\mu_j + \sigma_j^2/2}$. This captures two empirical regularities: (1) recovery driven by multiplicative factors (severity $\times$ age $\times$ medical quality $\times$ rehabilitation), naturally generating log-normal outcomes; (2) right-skewed distribution reflecting that most players recover as expected while a minority experience prolonged absence.

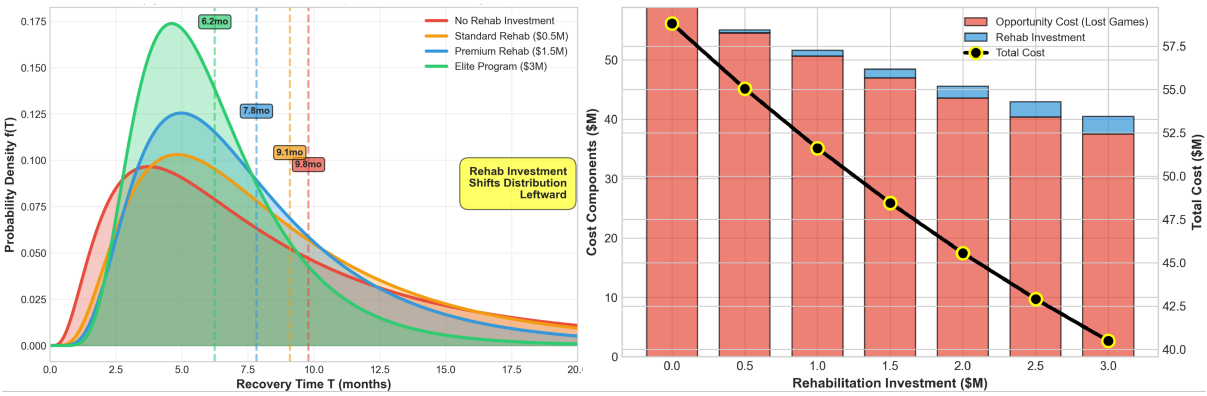


Figure 12: Rehabilitation Investment: Distribution Shift & Cost-Benefit Analysis

Under injury uncertainty, the MPC objective incorporates both expected value and variance penalty:

$$\max_{\mathbf{u}} \ \mathbb{E}[J(T)] - \sigma \cdot \text{Var}[J(T)] \tag{19}$$

where the value function $J(T)$ is explicitly partitioned by recovery timeline:

$$\begin{aligned}
J(T) = \Big(\sum_{k=t}^{t+T-1} \gamma^{k-t} + \sum_{k=t+T}^{t+H-1} \gamma^{k-t} \Big) \Big[(F_{k+1} - F_k) + \lambda (B_{k+1} - B_k) \\
+ \vec{\alpha}_1^T \mathbf{P}_{k+1} \cos\langle \mathbf{P}_{k+1}, \mathbf{P}^* \rangle \Big] + \Phi(\mathbf{S}_{t+H})
\end{aligned} \tag{20}$$

This formulation explicitly values the *option* embedded in rehabilitation investment: higher I_{rehab} shifts the recovery distribution leftward (reducing $\mathbb{E}[T]$), accelerating the transition from degraded to full roster strength.

Figure 12 shows that higher rehabilitation investment (I_{rehab}) shifts recovery time distributions leftward with reduced variance (panel a), significantly lowering prolonged absence risk. Panel (b) reveals that while direct costs increase, opportunity costs from lost games decrease substantially; the "Elite" \$3M strategy minimizes total cost for high-value players, validating aggressive medical intervention over conservative approaches.

8 Conclusions

This paper develops a hierarchical optimization framework that progressively addresses five interconnected challenges in sports franchise management. The **MPC foundation** (Task 1) formalizes the franchise as a state-space system tracking $[\mathbf{P}_t, B_t, F_t, D_t]$; the **RA-DUCB-MCTS algorithm** (Task 2) solves the $2^{n_{UFA}}$ combinatorial roster optimization, identifying Donovan Mitchell (23.07% skill enhancement) as the optimal acquisition; the **Bi-Level Strategic Equilibrium System** (Task 3) couples Nash expenditure games with MPC allocation, adaptively shifting toward cost-efficient acquisitions (Ingram at \$33M) under expansion pressure; and the extensions for **dynamic pricing** (Task 4) and **injury management** (Task 5) complete the unified six-dimensional control framework $[\mathbf{I}_r, I_{\text{marketing}}, I_{\text{debt}}, p, I_{\text{rehab}}, \mathbf{v}]$. In summary, our approach transforms franchise management into a *quantitative, adaptive, and constraint-aware* optimization system balancing "Winning" and "Profit" under uncertainty.

9 Model Evaluation

9.1 Strength

- **Hierarchical Modularity:** The bi-level architecture separates strategic budgeting (macro Nash game) from operational allocation (micro MPC), enabling seamless scenario analysis under structural shocks.
- **Multi-Objective Unification:** The single MPC objective function endogenously resolves the "Winning vs. Profit" trade-off without requiring ad-hoc weighting.
- **Constraint-Aware Feasibility:** NBA regulatory constraints and financial stability requirements are embedded as hard constraints, guaranteeing implementable strategies.

9.2 Weakness And Improvement

- **Parameter Calibration Uncertainty:** Key coefficients ($\vec{\alpha}_1$, β) rely on limited historical data. Future work should incorporate Bayesian estimation to generate robust confidence intervals.
- **Simplified Player Interactions:** The additive assumption ($\mathbf{P}_t = \sum_j \mathbf{P}_t^j$) neglects chemistry effects. Extending to pairwise interaction terms would better capture non-linear roster dynamics.

Letter to NBA Team New York Knicks

To: Team Owner and General Manager
From: ICM Team #2603012
Date: February 1, 2026
Re: *A Win-Win Strategy for Competitive Excellence and Financial Prosperity*

Dear New York Knicks Leadership,

As your dedicated modeling team, we present a unified strategy rooted in a core belief: **sports management thrives not on zero-sum trade-offs, but on ecosystems where players, fans, fellow teams, and ownership all prosper**. This framework weaves together data-driven decisions to balance on-court success with long-term financial health, tailored to the **New York Knicks'** unique needs.

► **2025-26 Season Strategy.** Our dynamic control system identifies **Donovan Mitchell** as the optimal acquisition—signing him at **$42 million annually (93% of available cap space)** will boost our Shooting Guard position strength by **23.07%**, closing our critical gap against championship benchmarks. His star power, paired with a **$16 million marketing investment**, will amplify brand equity and drive revenue across tickets, merchandise, and sponsorships, supported by a modest **$0.5 million in debt financing (<1% of brand equity)** to preserve liquidity.

► **League Expansion Response.** League expansion, far from a threat, becomes a catalyst for collective growth that attracts non-sports capital and broader media attention to the league. Our bi-level model shows that a "blue-ocean" expansion (e.g., Seattle) **increases our optimal budget by 3%**, enabling aggressive talent upgrades, while a direct competitor scenario (e.g., nearby market entry) prompts a pivot to cost-efficient acquisitions like **Brandon Ingram ($33 million)**—freeing **$4 million** for defensive marketing to protect our fan base while still delivering **21% performance improvement**. Even small-market expansions (e.g., Louisville) are met with targeted efficiency tweaks, ensuring we adapt without structural upheaval.

► **Dynamic Ticket Pricing.** Our strategy balances immediate revenue with fan loyalty: **$142 per ticket** for the upcoming contention season maintains **91.5% attendance**, while rebuilding phases keep prices accessible ($85) to nurture long-term season-ticket holders. Post-expansion, a modest reduction to **$118** defends market share without alienating supporters, recognizing fans as partners in sustainable growth rather than mere consumers.

► **Injury Management.** When key players face injury, our adaptive framework prioritizes win-win outcomes: **$3 million in rehabilitation investment** for high-value stars shortens recovery time (modeled via log-normal distribution) and preserves our competitive window, while balancing roster continuity with financial prudence. This approach not only gets players back on the court faster but fosters the loyalty that attracts top talent and strengthens team chemistry.

► **Risk Mitigation.** Naturally, trade-offs exist—concentrating cap space on star talent reduces short-term flexibility, and expansion brings market uncertainty—but our model mitigates risks through cost-efficient supporting casts, scenario-specific adjustments, and strict financial safeguards (debt limits tied to brand equity, interest-coverage requirements). Every decision aligns competitive success with financial sustainability: *winning drives revenue, and sound finances fund sustained talent investment*, creating a virtuous cycle that benefits all stakeholders.

We stand ready to refine this strategy with your guidance, confident it will position the Knicks as a perennial contender and a valuable, resilient franchise for seasons to come.

Respectfully submitted,
ICM Team #2603012

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Report on Use of AI

1. Google Gemini Pro

   Query1: What is Monte Carlo Tree Search?

   Output: MCTS stands for Monte Carlo Tree Search. It is a heuristic search algorithm used in decision-making processes, most famously in AI for games like Chess, Go (e.g., AlphaGo), and complex strategy simulations.

   - It works by building a search tree through four repeating steps:
   - Selection: Starting at the root, the algorithm selects child nodes based on a balance of exploitation (choosing known high-reward paths) and exploration (trying less-visited paths).
   - Expansion: Unless the current node ends the game, it creates one or more reachable child nodes.
   - Simulation (Playout): It performs a random simulation from the new node to a terminal state (a win, loss, or draw).
   - Backpropagation: The result of the simulation is passed back up the tree to update the statistics (win rates) of the nodes, informing future selections.

2. Google Gemini Pro

   Query1: Please write a very simple function implementation in MCTS for me.

   Output: Sure!

```python
def best_child(node, exploration_weight=1.41):
    """
    Selects the child node with the highest UCB1 score.
    """
    best_score = float('-inf')
    best_sub_nodes = []

    for child in node.children:
        # Exploitation term: average reward
        exploitation = child.wins / child.visits
        # Exploration term: encourages visiting less-explored nodes
        # math.log(node.visits) is the 'ln N' from the formula
        exploration = exploration_weight * math.sqrt(
            math.log(node.visits) / child.visits
        )
        ucb_score = exploitation + exploration
        if ucb_score > best_score:
            best_score = ucb_score
            best_sub_nodes = [child]
        elif ucb_score == best_score:
            best_sub_nodes.append(child)

    return random.choice(best_sub_nodes)
```